Doeun Kim

dragonhope@naver.com | Seoul, South Korea

Education

- **Sogang University** *Mar 2021 – Present*
  B.S. in Mathematics and Computer Science Seoul, South Korea
  - **GPA:** 4.16 / 4.30; **Current:** 7th semester
  - **Leave of Absence:** Mandatory military service, Oct 2022 – Apr 2024

Research Interests

Causal Machine Learning, Causal Foundation Models, Graph Representation Learning, Natural Language Processing

Research Experience

- **Causality Lab, SNU Graduate School of Data Science** *Jul 2025 – Present*
  Undergraduate Researcher (Advisor: Prof. Sanghack Lee) Seoul, South Korea
  - Investigated failure modes of Causal Foundation Models (CFMs) under post-treatment conditioning, query misspecification, and causal structure ambiguity.
  - Studied graph-aware approaches for incorporating structural information into PFN-style causal foundation models.
  - Co-authored ICML 2026 workshop work on CFM robustness and query specification.
- **SKI-ML Lab, SNU Graduate School of Data Science** *Jun 2024 – Aug 2024*
  Intern (Advisor: Prof. Jay-Yoon Lee) Seoul, South Korea

Teaching & Service Experience

- **Sogang University** *Winter 2024, Summer 2025*
  Teaching Assistant, Introduction to AI Programming (COR1010) Seoul, South Korea
  - Assisted course operation, programming assignment support, and student Q&A for an introductory AI programming course.
- **KATUSA, Republic of Korea Army** *Oct 2022 – Apr 2024*
  Military Service, Korean Augmentation to the U.S. Army South Korea
  - Completed mandatory military service in a bilingual U.S.-ROK Army environment.

Workshop Papers

[1] **A Structural View of Query Misspecification in Causal Foundation Models.** *ICML 2026 Workshop on Structured Probabilistic Inference & Generative Modeling (SPIGM), 2026.*
Causal Foundation Models Perform Better without Post-treatment Variables. *ICML 2026 Workshop on Foundation Models for Structured Data (FMSD), 2026.*

Projects

- **Do-PFN Really Do Causality?** *Fall 2025*
  - Collaborated with Causality Lab interns on a research project examining whether Do-PFN/PFN-style models exhibit causal reasoning behavior.
- **FA-AL-MOT: Multimodal Fashion Price Prediction** *Fall 2025*
  - Built a multimodal AI model for C2C fashion resale price prediction and product attribute classification using image and text features.
  - Used CLIP fusion, LLM-based attribute restoration, and GPU-accelerated deep learning; achieved $R^2 = 0.8904$ for price prediction.
- **DIGIRO: Real-Time Notification Service** *Summer 2025*
  - Architected and developed a real-time notification service using FastAPI (Python) and PostgreSQL to aggregate and serve live data for Seoul's Jongno-gu district.

- Containerized the application's backend, crawler, and database services using Docker and managed deployment orchestration with Kubernetes for scalability and reliability.
- Integrated multiple external sources, including Seoul Public Data and Kakao Maps APIs, to provide users with an interactive map visualizing protest locations, routes, and traffic control areas.

- **News Summarization with GPT-2 & Knowledge Distillation** *Spring 2025*
  - Enhanced GPT-2 summarization on CNN/DailyMail dataset; sequence-level distillation from LLaMA3-8B improved ROUGE-1 by 20.6% and METEOR by 54.6%.
  - Analyzed decoder-only model limitations and cross-model distillation challenges.
- **Convenience-Store Management Application** *Spring 2025*
  - C/MySQL CLI tool for business analytics queries.
- **Automated Lighting Control via Object Detection** *Winter 2024*
  - Integrated YOLO + ByteTrack with a line-crossing counter to trigger lights based on occupancy.
- **RAG Chatbot for Academic Catalog** *Fall 2024*
  - Built a RAG pipeline with SOLAR embeddings, SOLAR API, ChromaDB, and Streamlit UI for real-time Q&A on Academic Catalog.
- **Fine-Grained Classification** *Spring 2024*
  - Developed ResNet-101 classifier on CUB-200-2011; augmented data with Gaussian noise to defend against PGD attacks.

Scholarships

- **KT Digital Talent Scholarship**, KT (Full Funding) Spring 2025 – Present

Awards & Honors

- **Outstanding Scholar**, KT Group Hope Sharing Foundation (selected as one of 4 outstanding scholars) Jan 2026
- **Excellence Award in Inquiry Community Program (for *Automated Lighting Control via Object Detection*)**, Sogang University Jan 2026
- **Dean's List**, College of Natural Sciences, Sogang University (Year of 2021) 2022

Skills

- **Programming & Tools:** C/C++, Python, MySQL, SAS, MATLAB
- **Languages:** Korean (Native), English (Fluent)